Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Elements of which of the following group(s) of periodic table do not form hydrides?

(A) Groups $7, 8, 9$

(B) Group 13

(C) Groups $15, 16, 17$

(D) Group 14

Q.2

Among CaH_2, NH_3, NaH and B_2H_6 which are covalent hydrides?

(A) NH_3 and B_2H_6

(B) NaH and CaH_2

(C) NaH and NH_3

(D) CaH_2 and B_2H_6

Q.3

The alkali metals form salt-like hydrides by the direct synthesis at elevated temperature. What is the correct order of thermal stability of these hydrides ?

(A) NaH > LiH > KH > RbH > CsH

(B) LiH > NaH > KH > RbH > CsH

(C) CsH > RbH > KH > NaH > LiH

(D) KH > NaH > LiH > CsH > RbH

Q.4

Which of the following chemicals is not present in clear hard water?

(A) $MgCO_3$

(B) $MgSO_4$

(C) $Mg(HCO_3)_2$

(D) $CaCl_2$

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Q.5

Choose the correct statement:
The reason for use of polyphosphates as water softening agents is, that

(A) they form soluble complexes with anionic species

(B) they precipitate out cationic species

(C) they precipitate out anionic species

(D) they form soluble complexes with cationic species

Q.6

Which of the following is not true?

(A) D_2O freezes at lower temperature than H_2O

(B) Reaction between H_2 and Cl_2 is much faster than D_2 and Cl_2

(C) Ordinary water gets electrolysed more rapidly than D_2O

(D) Bond dissociation energy of D_2 is greater than H_2

Q.7

Hard water is not fit for washing clothes because

(A) it contains Na_2SO_4 and KCl

(B) It gives precipitate

(C) it contains impurities

(D) it is acidic in nature

Q.8

Which of the following statements do not define the characteristic property of water "Water is a universal solvent"

(A) It can dissolve maximum number of compounds

(B) It has very low dielectric constant

(C) It has high liquid range

(D) None of these

Q.9

Chemical A is used for water softening to remove temporary hardness. A reacts with sodium carbonate to

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generate caustic soda. When CO_2 is bubbled through a solution of A, it turns cloudy. What is the chemical formula of A?

(A) $CaCO_3$

(B) CaO

(C) $Ca(OH)_2$

(D) $Ca\,(HCO_3)_2$

Q.10

Which physical property of dihydrogen is wrong?

(A) Odourless gas

(B) Tasteless gas

(C) Colourless gas

(D) Non-inflammable gas

Q.11

Permanent hardness in water cannot be cured by:

(A) Treatment with washing soda

(B) Boiling

(C) Calgon's method

(D) Ion exchange method

Q.12

The alum used for purifying water is

(A) Ferricalum

(B) Chromealum

(C) Potash alum

(D) Ammonium alum

Q.13

What is formed when calcium carbide reacts with heavy water?

(A) C_2D_2

(B) CaD_2

(C) Ca_2D_2O

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(D) CD_2

Q.14

The dielectric constant of H_2O is 80. The electrostatic force of attraction between Na^+ and Cl^- will be

(A) reduced to $1/40$ in water than in air

(B) reduced to $1/80$ in water than in air

(C) will be increased to 80 in water than in air

(D) will remain unchanged.

Q.15

The numbers of protons, electrons and neutrons in a molecule of heavy water are respectively:

(A) $8, 10, 11$

(B) $10, 10, 10$

(C) $10, 11, 10$

(D) $11, 10, 10$

Q.16

Which of the following terms is not correct for hydrogen?

(A) Its molecule is diatomic

(B) It exists both as H^+ and H^- in different chemical compounds

(C) It is the only species which has no neutrons in the nucleus

(D) Heavy water is unstable because hydrogen is substituted by its isotope deuterium

Q.17

Which of the following is a salt-like hydride?

(A) NH_3

(B) H_2O

(C) NaH

(D) $TiH_{1.73}$

Q.18

Ionic hydrides are usually

(A) liquids at room temperature

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(B) good reducing agent

(C) good electrical conductors in solid phase

(D) easily reduced

Q.19

Why is D_2O preferred over H_2O as a moderator in nuclear reactors?

(A) D_2O slows down the speed of neutrons more effectively than H_2O.

(B) D_2O has high specific heat.

(C) D_2O is cheaper.

(D) None.

Q.20

Which of the following statements is incorrect?

(A) Hardness of water is shown by its behaviour towards soap.

(B) The temporary hardness is due to the presence of Ca and Mg bicarbonates.

(C) Permanent hardness is due to the presence of soluble Ca and Mg sulphates and chloride.

(D) Permanent hardness can be removed by boiling the water.

Answers & Solutions

Q.1 Answer:

Groups $7, 8, 9$

Solution:

Elements of group 7,8 and 9 of d-block do not form hydrides at all. The inability of metals of group 7,8 and 9 of periodic table to form hydrides is referred to as hydrides gap of d-block. In these compounds H atoms are supposed to occupy interstitial position in the metal lattices. They are also called nonstoichiometric hydrides.

Q.2 Answer:

NH_3 and B_2H_6

Solution:

N and B belong to p-block and since the electronegativity difference between these elements and H is very less, they form covalent hydrides.

Q.3 Answer:

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LiH > NaH > KH > RbH > CsH

Solution:

The alkali metal hydride stability $\propto$ lattice energy

On moving top to bottom the lattice energy decreases hence, the thermal stability of hydrides decreases.

Q.4 Answer:

$MgCO_3$

Solution:

$MgCO_3$ is insoluble in water.

Q.5 Answer:

they form soluble complexes with cationic species

Solution:

Polyphosphates (e.g., sodium hexa metaphosphate) form soluble complexes with cations such as Ca^{2+}, Mg^{2+} and so they are used as water softening agents.

$$Na_2[Na_4(PO_3)_6] + 2Ca^{2+} \longrightarrow Na_2[Ca_2(PO_3)_6] + 4Na^+$$

"Calgon" Hardness "Soluble" Sod. hexa metaphosphate

Q.6 Answer:

D_2O freezes at lower temperature than H_2O

Solution:

D_2O actually has higher freezing point $(3.8^\circ C)$ than water H_2O $(0^\circ C)$

Q.7 Answer:

It gives precipitate

Solution:

$$2RCOONa + Mg^{2+} \rightarrow (RCOO)_2Mg + 2Na^+$$

Insoluble

$$2RCOONa + Ca^{2+} \rightarrow (RCOO)_2Ca + 2Na^+$$

Insoluble

Q.8 Answer:

It has very low dielectric constant

Solution:

Water has high dielectric constant i.e., $78.39\ C^2/Nm^2$, high liquid range and can dissolve maximum number of compounds. That is why it is used as universal solvent.

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Q.9 Answer:

$Ca(OH)_2$

Solution:

$Ca(OH)_2$ **is used for the softening of temporary hard water.**

$Ca(OH)_2(aq) \xrightarrow{CO_2} CaCO_3(s) + H_2O(l)$

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Q.10 Answer:

Non-inflammable gas

Solution:

H_2 **is a highly inflammable gas.**

Q.11 Answer:

Boiling

Solution:

Only temporary hardness which is due to HCO_3^- **(bicarbonate) ions is removed by boiling.**

Q.12 Answer:

Potash alum

Solution:

Potash alum is used for purifying water.

Q.13 Answer:

C_2D_2

Solution:

$$CaC_2 + 2D_2O \rightarrow C_2D_2 + Ca(OD)_2$$

Q.14 Answer:

reduced to $1/80$ in water than in air

Solution:

Electrostatic forces of attraction are reduced to $1/80$ **th in water.**

Q.15 Answer:

$10, 10, 10$

Solution:

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Heavy water is D_2O hence number of electrons $= 2 + 8 = 10$
number of protons $= 10$ Atomic mass of $D_2O = 4 + 16 = 20$
hence number of neutron
= Atomic mass − number of protons

$$= 20 - 10 = 10$$

Q.16 Answer:

Heavy water is unstable because hydrogen is substituted by its isotope deuterium

Solution:

Heavy water is stable.

Q.17 Answer:

NaH

Solution:

Hydrides of alkali metals are salt like.

Q.18 Answer:

good reducing agent

Solution:

good reducing agent

Q.19 Answer:

D_2O slows down the speed of neutrons more effectively than H_2O.

Solution:

D_2O **slows down the speed of neutrons more effectively than** H_2O**.**

Q.20 Answer:

Permanent hardness can be removed by
boiling the water.

Solution:

Permanent hardness can be removed by
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